

TECHNICAL SPECIFICATION FOR OUTDOOR TYPE DISTRIBUTION PLOT & SERVICE JUNCTION BOX

1. Scope:

This specification covers technical requirements of design, manufacture, testing at manufacturer's works, packing, forwarding, supply and unloading at store/site and performance of Plot & Service junction box with all accessories for trouble free and efficient operation. This outdoor Plot & Service junction box will be utilised for providing/interconnecting service main of distribution network.

2. Applicable Standards:

The equipment covered with latest editions of the following standards/IEC:

- a) **IS 5039-1983:** Specification for distribution pillars for voltages not exceeding 1000V AC & 1200V DC.
- b) **IS 13703 (Part-1&2)-1993 / IEC 60263/1-1986:** Specification for LV fuses for voltage not exceeding 1000V AC & 1500V DC.
- c) **IS 12063-1987 / IEC 60529:** Classification of degree of protection provided by enclosure of electrical equipment.
- d) **IS 2551-1982:** Danger Notice Plates.
- e) **IEC 60664:** Insulation co-ordination within low voltage systems including clearance creepage distances for equipment.
- f) **IS 2147:** Enclosure & Degree of protection of the Plot & Service junction box
- g) **IS 6005:** Code of Practice for phosphating iron and steel

3. Climatic Conditions of the Installation:

- a) Max. Ambient Temperature : 42 deg. C
- b) Min. Ambient Temperature : 10 deg. C
- c) Max. Humidity : 98%
- d) Min. Humidity : 60%
- e) Average Annual Rainfall : 2500 mm
- f) Average No. of rainy days per annum : 150 days
- g) Altitude above MSL : 9 metres
- h) Average wind velocity : 18 km/hr.

4. General Technical Requirement:

Sl. No.	Description	Plot Junction Box	Service Junction Box
1.	Installation	Outdoor	Outdoor
2.	Rated Voltage	430V	430V
3.	No. of Phases	3	3
4.	Frequency	50 Hz	50 Hz
5.	Wiring System	3 Phase 4 Wire	3 Phase 4 Wire
6.	System Earthing	Solid at source end.	Solid at source end.
7.	Current Rating (A)	400 A	200 A
8.	Dimension(H x W x D)	Specified by vendor(Foundation plan attached in Annexure A)	Specified by vendor(Foundation plan attached in Annexure A)
9.	Busbar Material/Size		
	Phase (3 Nos)	Cut out (400A) - 3 nos	EC Grade Aluminium (50mm x 8 mm)
	Neutral (1 No)	EC Grade Aluminium (60 x 10 mm)	EC Grade Aluminium (50mm x 8 mm)
10.	Max. current density of AlBusbar	1 Amp/sqmm.	1 Amp/sqmm.
11.	Degree of Protection	IP-43	IP-43
12.	Temperature Rise	Max. permissible temperature rise for the Busbar and terminal should be 45°C & 65°C respectively at ambient not exceeding 42°C	Max. permissible temperature rise for the Busbar and terminal should be 45°C & 65°C respectively at ambient not exceeding 42°C
13.	No. of Incoming	2	2
14.	No. of Outgoing	NA	1. 6 Nos. (100A cut-out) for 3-phase connection (2 consumer) 2. 6 Nos. (32A cutout) & 3 Nos. (63A cutout) for single phase connection.
15.	Fuse Base Rating & Size	400 A	32A , 63 A ,100A
16.	Current rating of Busbar	600A	400A
17.	Busbar spacing (centre to centre)	Specified by vendor	Specified by vendor
18.	Neutral Busbar rating	600A	400A
19.	Size of Incoming Cables	4X 1C X 400 sqmm., 4C X 185 sqmm.	1X 4C X 185sqmm.

20.	Size of Outgoing Cables	NA	1X 4C X 50sqmm., 4C X 25, 4C X 16(for 3-phase) 2CX6, 2Cx10,2Cx16,2Cx25,2Cx50(For single phase supply)
21.	Box Leg dimension	Specified by vendor	Specified by vendor
22.	Enclosure material	Cold rolled steel of 2.5 mm thickness	Cold rolled steel of 2.5 mm thickness
23.	Colour of enclosure	Light Grey 631 of IS:5	Light Grey 631 of IS:5
24.	Cable Entry/Cable Gland	Bottom/3 mm Cable Gland	Bottom/3 mm Cable Gland

5. General Construction:

- a) The design and layout of Plot & Service junction box shall be compact and convenient to provide ease of termination of cables and operation of fuse bases. It shall be designed for termination of cables and operation of fusebase shall be from the front end. The Plot & Service junction box enclosure shall provide a degree of protection not less than IP-43 in accordance with IS-2147.
- b) The enclosure shall be suitable for Outdoor installation. The enclosure shall be made from cold rolled sheet steel of thickness 2.5 mm and shall be suitable to withstand rough usage and weather.
- c) Box shall be rigid free standing & floor mounting type and comprise of structural frames enclosed completely with specially selected smooth finished cold rolled sheet steel of thickness not less than 2.5 mm for weight bearing members of the panels such as front sheet and door frames, sides, door top and bottom portions.
- d) All joints to be made flush type and all edges shall be bent at right angles and rounded. All structural members shall be bolted or welded together. Necessary arrangement shall be provided for bolting together the adjacent panels. The doors, removable covers and panels shall be sealed with gasket all round with neoprene gaskets. Ventilating louvers shall be provided at both sides & shall have screens and filters.
- e) The enclosure shall be sized to facilitate easy maintenance at minimum height of 300 mm above FFL.
- f) The top of pillar shall be fitted with a sloping canopy, the design of which shall be such that rain water shall not accumulate on the top.
- g) It shall have overhang and fascia of min. 100mm. 2 nos. lifting lugs of same material shall be provided.
- h) Three nos. heavy duty hinges shall be provided for each door such that they are not visible from outside and hence not removable. The hinge shall be of such construction that the door can be swung open by not less than 150°. The door shall be fitted with integral handle, flushed with the surface of the door.

- i) Air ventilation arrangement shall be provided at top and all sides of the enclosure to facilitate air circulation.
- j) The door shall have three point locking arrangement. Special type tamperproof heavy duty built-in inter-lock with appropriate key shall be provided. Locking system with handle up & down arrangement shall be provided on front side of the doors. The door shall be provided with pad-locking facility also.
- k) Proper support shall be provided for holding I/C& O/G cables. These, cable support arrangement shall be provided to take the load of the cable.
- l) The Busbar of all phases and neutral shall be provided with colour coded heat shrinkable insulation sleeves.
- m) Cable termination arrangement shall be from bottom and suitable for external cables of type and sizes as mentioned in the specification.
- n) The box shall be provided with an Aluminium Earth bus at the bottom, suitable for the rated short time current of the box. The earth bus shall provide Earthing to all the components inside the pillar and the metallic enclosure including the armour of the cables. The earth bus shall be internally connected to two separate earth terminals located outside, on either side of the Box.
- o) The paint on the Plot & Service Junction box shall have good colour retention properties and shall be weather and UV resistant. The Retro reflective paint strip shall be provided on the four faces of the pillar box which shall withstand the temperature conditions and reflect vehicular light at night time providing safety to traffic. All sheet steelwork shall be phosphated in accordance with IS: 6005 "Code of practice for phosphating iron and steel". Oil, grease, dirt and swart shall be thoroughly removed by emulsion cleaning. Rust and scale shall be removed by pickling with dilute acid followed by washing with running water rinsing with slightly alkaline hot water and drying. After phosphating, thorough rinsing to be carried out with clean water followed by final rinsing with diluter dichromate solution. After that the powder spray with spray gun to be used and to put into the oven for drying in 200deg.C. for 40min
- p) The Plot & Service Junction box shall be provided with durable and legible Aluminium nameplates containing all technical parameters and manufacturing details. A plate of statutory instruction like 'stick no bill' shall be provided on all four side of the box. The box shall be provided with a plate 'Danger' mark for 430V AC as per IS 2551-1982. Property plate mentioning "Property of WBSEDCL" and project name if any shall also be provided.
- q) All routine, acceptable & type tests shall be carried out in accordance with the relevant IS/IEC. All routine/acceptance tests shall be witnessed by the Purchaser/his authorized representative.
- r) Equipment shall be subject to pre-dispatch inspection by a duly authorized representative of the Purchaser. Inspection may be made at any stage of manufacture at discretion of the

purchaser and the equipment, if found unsatisfactory as to workmanship or material, the same is liable to rejection.

- s) The material received at Purchaser store will be inspected for acceptance and shall be liable for rejection, if found different from the reports of the pre-dispatch inspection.
- t) Supplier shall stand guarantee for five years from the date of supply towards design, materials, workmanship and quality of process/manufacturing of items under this contract.
- u) Supplier shall ensure that all the equipment covered under this specification shall be prepared for transport in a manner so as to protect the equipment from damage in transit.
- v) Supplier shall submit with the offer Quality assurance plan indicating the various stage of inspection, the tests and checks which will be carried out on the material of construction, components during manufacturing and equipment after finishing.
- w) 6 nos. holes of suitable size shall be drilled on the busbar/ phase of the Plot & Service Junction boxes.
- x) Holes for cable entry at bottom shall be as follows:

PLOT JB	1. 8 nos. (4 nos. X 2) suitable size for 1 C X 400 sqmm 1.1 KV XLPE cable.
	2. 2 nos. suitable size for 4 C X 185 sqmm 1.1KV XLPE cable
SERVICE JB FOR THREE (3) PHASE CONNECTION	1. 2 nos. suitable size for 4C X 185sqmm XLPE cable
	2. 2 nos. suitable size for 4C X 185 / 4C X 50 / 4 C X 25 / 4C X 16 sqmm XLPE cable
SERVICE JB FOR SINGLE (1) PHASE CONNECTION	1. 2 nos. suitable size for 4C X 185sqmm XLPE cable
	2. 9 nos. suitable size for 4C X 50 / 4 C X 25 / 4C X 16 sqmm XLPE cable

6. Special Requirement:

Wider Spacing at terminals for connecting cable so that it is easy for operators to connect /disconnect cables.

7. TESTS:

7.1 TYPE TESTS: ON COMPLETE PLOT & SERVICE JUNCTION BOX

The equipment offered in the tender should have been successfully type tested at NABL laboratories in India or equivalent international laboratories in line with the relevant standard and technical specification, within the last 5 (five) years from the date of offer. The bidder shall be required to submit complete set of the type test reports along with the offer.

The list of type tests is as follows:

Following type tests as per IS: 5039/1983 shall be carried out on complete L.T. Plot & Service junction Box.

1. Temperature rise test:-The temperature rise test should be carried out as per IS: 8623 –1993.
2. High voltage test shall be carried out as per IS: 8623/ 1993 amended up to date.
3. Short Time Withstand Current Test on L.T. Plot & Service junction Box shall be carried out as per IS 8623: or latest version.
4. Degree of protection for IP- 43 on complete L.T. Plot & Service junction Box shall be carried out as per IS: 13947/1993 or the latest version thereof.
5. Verification of clearance & creepage distances.

7.2 ACCEPTANCE TEST: ON COMPLETE PLOT & SERVICE JUNCTION BOX

Following tests shall be carried out as per acceptance tests in addition to routine tests on one random sample of each rating out of the lot offered for inspection:

- i) Temperature rise test on one sample of each rating.

7.3 ROUTINE TESTS: ON COMPLETE PLOT & SERVICE JUNCTION BOX

- a) Routine and acceptance tests as per relevant IS shall be carried out at the manufacturers' works.
- b) Overall Dimensions Checking.
- c) Insulation Resistance Tests.
- d) High Voltage Test at 2500 V, 50 Hz AC for one minute.

8. INSPECTION:

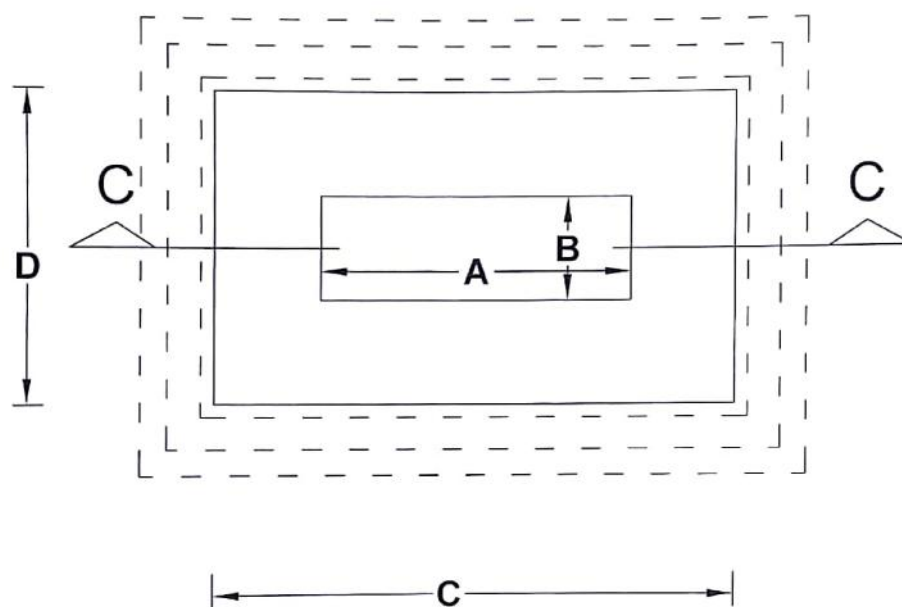
- 8.1. The supplier shall prepare a prototype and offer the same for inspection and approval before taking up mass production.
- 8.2. All routine & acceptance tests and inspection of material shall be carried out at the place of manufacturer. The manufacturer shall offer the Inspector (representing the purchaser) all reasonable facilities, free of charge at the time of Inspection.
- 8.3. The supplier shall offer inspection of the material at his works before dispatch. If required the supplier shall also offer inspection of manufacturing painting and assembly processes and quality control system. If any material is not found in line with relevant specification the supplier shall carry out the modification and offer for re-inspection.
- 8.4. Inspection and acceptance shall not absolve the supplier of his responsibility to supply the material in accordance with the specifications. The purchaser reserves the right to reject the material not confirming the relevant specifications.

9. DRAWINGS AND DOCUMENTATION:

- a. The successful bidder shall submit set of all drawings of the Plot & Service junction Boxes and its components intricate and get the same approved before commencement of production.
- b. The tenderer shall furnish all details and clarifications required if any for scrutiny and evaluation of the offer.
- c. Manufacture of material to be supplied shall be done strictly as per approved drawing.
- d. Approval of drawing shall not absolve the supplier of his liability for ensuring correctness according to applicable standards & regulations.

10. DESPATCH:

- 10.1 The material to be supplied shall be packed and dispatched only after inspection and approval.
- 10.2 Supplier shall be responsible for packing, transporting and delivery to the consignee at anywhere of West Bengal.
- 10.3 Copies of packing list and inspection report duly approved shall be sent along with each consignment.

ANNEXTURE - APLAN OF BLOCK/ PLOT J.B. AND
SERVICE J.B. FOUNDATION

For	A	B	C	D
Plot/ Block JB	600	200	1000	600
Service JB	400	100	800	500